Tell

# Your contract hides the vote. The ledger still shows it.

A privacy leak scanner for Midnight Compact contracts: what each circuit gives away, in bits, with a fix Tell has compiled and checked.

Midnight Buildathon · Wave 3 · November 2026



# The compiler says whether. Tell says how much.

This line compiles. disclose() is there.

```
assert(c < 4, "four candidates");
ballots.insert(tag,
disclose(persistentHash(c)));
```

Midnight keeps the vote out of the transaction. But the ledger stores its hash, and there are only four choices.

What Tell reports

## 2 bits

Hash 0, 1, 2 and 3, compare with the ledger, read every vote. Grade F, critical, with the line, the public transcript and a fix that has been compiled and re-scanned.

# Proved on the chain, not just on paper

**6 / 6**

votes read back from the leaky ballot on Midnight Preprod, by the page you can open right now

**0 / 6**

from the fixed ballot, with the same six votes cast: the fix Tell suggests, deployed

tell-midnight.vercel.app/live reads both contracts from the indexer. A daily GitHub workflow checks the same numbers against the live site.

Wave 2: 20 points, tied for the top · Engineering 8 · Product 8 · QA 7 · UX 7 · Communication 7 · Business 6

# You said, we did

| | |
|---|---|
| “Deploy to testnet and add the contract address to your README.” | Done: both ballots live with addresses in the README. Then a second contract of our own, Tell Desk, deployed on Preprod. |
| “Please consider applying to our technical fellowship program.” | Applied to the Aliit fellowship (Builder and Educate). Posted Tell as input to Midnight’s MPS-0036 on review evidence. |
| “Commercial relevance and viability will be a central focus … even if the numbers are small.” | Notes to 11 teams, a fix offered to Midnight’s own examples, two audit firms asked for a pilot, and a priced model with its ceiling stated. |

# “As a business it is small.” Each substitute, answered

| Substitute you named | What Tell does that it does not |
|---|---|
| The Compact compiler | disclose() says whether data goes public; Tell measures how much, per circuit, in bits |
| Audit firms (Trail of Bits, Least Authority) | Runs before them and on every pull request; gives them a privacy map to start a review |
| Slither, Semgrep, Snyk | They do not read Compact or zero-knowledge transcripts |
| Nightshade, compact-guard, OpenZeppelin | Benchmarked head to head: Tell finds 23 of 25 planted leaks with 1 false alarm |
| “A few dozen teams” | Buyers are audit firms and the Foundation, each serving many teams; seats come last |

# New in Wave 3

### Ledger 9 and Compact 0.36

Events and cross-contract calls are public sinks; compilers 0.30 to 0.36 in CI.

### Constructors and tokens

Deploy-time leaks, shielded coin nonces, wallet keys, private amounts.

### Verified auto-fix

A patch Tell compiles and scans again before offering it. Ballot F to B, vault F to A.

### Pull-request review

The Action comments on the leaking line with the bits revealed and the fix.

### Tell Desk

A Compact contract on Preprod: sealed reports and on-chain privacy grades.

### Where developers work

VS Code extension, an MCP server for AI agents, Radar for deployed contracts.

# Tell Desk: tell the team before you tell the world

A Compact contract on Midnight Preprod, its whole lifecycle run: 17 transactions, 4 sealed reports, a finding acknowledged, fixed and revealed, 4 grades.

### On the public ledger

- A commitment to each report, never its text or project
- A researcher tag that differs on every report
- Status: pending, acknowledged, fixed, revealed
- Grades keyed by a hash of the verifier keys

### In private state, as proofs

- The report, its project and its nonce
- The researcher’s secret
- The maintainer’s credential, proved against a Merkle tree
- So a maintainer answers without saying which project

# On the pull request, with a fix that compiles

Tell writes the patch, compiles it with the same compiler, scans it again, and offers it only if the finding is gone and nothing new appears. Changes that alter behaviour are labelled.

**F** → **B** and the zoo attack test stops working

Live example: github.com/RohanGlitched/tell/pull/1

**Tell · critical · guessable hash**

2 bits of choice revealed; at most 4 guesses recover it.

```
- ballots.insert(tag,
- disclose(choiceHash(c)));
```

```
+ ballots.insert(tag, disclose(
+ persistentCommit(c, ballotSalt())));
```

Verified: compiles with 0.31.1, finding gone

# How Tell compares, head to head

| Tool, on the same 25 planted leaks | Found | False alarms | Measured in bits |
|---|---|---|---|
| Tell | 23 / 25 | 1 (0 with tell.json) | yes |
| Nightshade, secrets policy | 14 / 25 | 12 | rarely |
| Nightshade, default rules | 3 / 25 | 0 | rarely |
| compact-guard | 5 / 25 | 1 | by domain size |
| compact-lint, OpenZeppelin | 0 / 25 | 0 | no |

On Nightshade's own published corpus Tell analysed 443 of 465 contracts with no crashes: 30 findings at medium or worse, 23 real when labelled by hand.

Caveat, stated in bench/RESULTS.md: we wrote the planted cases; the corpus run and Midnight's examples are the less biased evidence.

# Tested like a product

**407**

tests in six workspaces, up from 97 in Wave 2

**28**

zoo tests: attacks that read a secret from public state, and fail on the fix

**5**

Compact compilers in CI, 0.30 to 0.36

**daily**

a workflow checks the live site against the chain

CI also builds the website and runs its browser and API tests, compiles Tell Desk under 0.35 and 0.36, and builds its proving keys for an offline dry run of the whole lifecycle.

# Radar: scanning Preprod with no source code

Radar decodes every contract it can reach from public chain data and tries to reverse each stored hash against small domains, using Compact's own hashing.

**329**

contracts decoded; 178,098 state cells and 5,014 call transcripts

**84**

contracts where one value appears in every call: a sign their callers may be linkable

**6 / 6**

votes recovered from Tell's leaky ballot without its source; none anywhere else

Counts only are published. Per-contract results stay private and go to their owners.

# Taking it to market, so far

| | |
|---|---|
| **11** | buildathon teams told there is a finding in their repository, with the details offered privately |
| **2** | contributions to Midnight's own examples: an issue on the election example, with the fix offered, and a README pull request that states the leak |
| **2** | audit firms that review Compact asked for a pilot; no answer yet |
| **1** | proposal discussion on Midnight's MIP repository, offering Tell as a reference for MPS-0036 |

Replies and fixes are tracked live at tell-midnight.vercel.app/adoption, as small as they are.

# Why now

## Review is now self-service

Midnight retired its pre-deployment contract review in October. Its risk rubric, led by “Privacy-at-Risk”, is now a self-assessment.

## The gap is on record

MPS-0036 notes that successful builds give “no stable privacy-flow account”. Tell produces one.

## Ledger 9 widens the surface

Since November 6 on mainnet: events and cross-contract calls, new places for private data to escape.

## Regulators agree

The EDPB’s July 2026 blockchain guidelines say unsalted hashes of personal data are, in general, not enough.

# Who pays, and how big it honestly is

| Buyer | Offer | Price we will test |
|---|---|---|
| Audit firms | A privacy map for every Compact review | $3,000–6,000 a year |
| Midnight Foundation | Rules kept current with each compiler | $30,000–60,000 grant |
| Teams with private code | The pull-request bot, hosted | $49 a repository a month |
| Public repositories | Everything, open source | Free, always |

Five firms, one grant and about 14 production teams put the ceiling near **$53,000–98,000 a year**. Our year-one target is $15,000–40,000, mostly grant and audit work. The same leak class exists on Aztec and Aleo, and that is the path past it.

Prices are hypotheses for pilots. Market figures and sources: BUSINESS-RESEARCH in the repository and /teams on the site.

# Next twelve months

**Dec – Feb**
One paid pilot or letter of intent with an audit firm; a grant request; a privacy-flow format proposed for MPS-0036

**Mar – May**
Hosted pull-request checks in beta; 20 repositories running the Action

**Jun – Nov**
Signed release reports; decide on Aztec from demand

## How Midnight can help

- Link a privacy check from the self-assessment rubric and project templates
- Keep a machine-readable privacy-flow output in the compiler
- Introduce us to the firms auditing Compact
- Review our pull request on the election example

# Every contract has a tell. Find it before the chain does.

tell-midnight.vercel.app

uses: RohanGlitched/tell@v1

github.com/RohanGlitched/tell

Apache-2.0 · Rohan Borade